

LEARNING ROAD-NETWORK RESILIENCE UNDER DISRUPTIONS WITH SPECTRAL GRAPH THEORY AND GRAPH NEURAL NETWORKS

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ABSTRACT. We estimate structural connectivity loss in spatial graphs under simultaneous edge disruptions. We compare a first-order Fiedler-vector approximation, a direct graph convolutional network (GCN), an ablation without the node-level Fiedler feature, and a residual spectral-GCN that learns a bounded correction to the analytical estimate. A graph-disjoint evaluation uses 10,000 synthetic scenarios over five seeds, followed by zero-shot transfer to 1,500 scenarios on a 163-node OpenStreetMap road network in Ho Chi Minh City. On synthetic test graphs, residual learning reduces mean absolute error from 0.0656 ± 0.0132 to 0.0390 ± 0.0099 and raises Spearman correlation from 0.6279 ± 0.0571 to 0.8764 ± 0.0375 . Removing the Fiedler node feature has only a small effect, showing that the improvement arises mainly from using the spectral estimate as a residual prior. On OpenStreetMap transfer, the residual model achieves 0.1018 ± 0.0155 MAE and 0.9269 ± 0.0119 Spearman correlation. The standalone first-order approximation fails under this sparse network shift, demonstrating both the value and limitation of the hybrid correction. This is a structural-disruption study, not a flood-risk model.

1. INTRODUCTION

Road-network disruptions caused by extreme weather, construction, or local failures can alter global connectivity even when only a small subset of links is unavailable. A common computational task is therefore to rank many possible disruption scenarios. Exact spectral recomputation is well defined, but a fast surrogate may become useful when the number of networks and scenarios is large.

This work connects two viewpoints. Spectral graph theory supplies a precise connectivity measure and an interpretable local sensitivity. Graph neural networks (GNNs) provide a data-driven mechanism for representing interactions among several failed edges. Instead of presuming that the learned model must dominate the mathematical approximation, we ask where each method is reliable.

Our contributions are:

- (1) a normalized prediction target comparable across graphs;
- (2) an explicit first-order Fiedler-vector baseline;
- (3) a small graph-level regressor requiring only standard PyTorch;
- (4) a graph-disjoint experimental split that prevents structural leakage;
- (5) a five-seed comparison stratified by disruption size;
- (6) a Fiedler-feature ablation, a residual hybrid, and zero-shot OSM validation.

2. MATHEMATICAL FORMULATION

Let $G = (V, E, w)$ be a connected, undirected graph with positive edge weights. Its weighted adjacency matrix is A , degree matrix is D , and combinatorial Laplacian is $L = D - A$. Write the ordered eigenvalues as

$$0 = \lambda_1(L) \leq \lambda_2(L) \leq \cdots \leq \lambda_{|V|}(L).$$

Fiedler [1] introduced λ_2 as the algebraic connectivity. It is positive exactly when an undirected graph is connected.

Definition 2.1 (Relative connectivity loss). *For a set of disrupted edges $S \subseteq E$, let $G - S$ denote the damaged graph. We define*

$$(1) \quad y(G, S) = \text{clip} \left(1 - \frac{\lambda_2(G - S)}{\lambda_2(G)}, 0, 1 \right).$$

Thus $y = 0$ represents no measured loss, while $y = 1$ includes disconnection.

Suppose λ_2 is simple and u_2 is its unit eigenvector. For edge $e = (i, j)$, standard symmetric eigenvalue perturbation gives

$$(2) \quad \frac{\partial \lambda_2}{\partial w_e} = (u_{2,i} - u_{2,j})^2.$$

Removing several edges is a finite, not infinitesimal, perturbation. We use the clipped first-order prediction

$$(3) \quad \hat{y}_{\text{spec}}(G, S) = \text{clip} \left(\frac{\sum_{(i,j) \in S} w_{ij} (u_{2,i} - u_{2,j})^2}{\lambda_2(G)}, 0, 1 \right).$$

This expression ignores higher-order interaction and can be inaccurate near an eigenvalue crossing or disconnection. Its advantages are transparency and the absence of learned parameters.

3. SCENARIOGCN

For each scenario, the model receives the normalized adjacency matrix of $G - S$, augmented with self-loops, and five node features: normalized degree in G , fraction of incident edges in S , absolute normalized Fiedler coordinate, and two spatial coordinates. Three graph convolution layers use the update

$$H^{(\ell+1)} = \text{ReLU} \left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(\ell)} W^{(\ell)} \right),$$

following the localized propagation principle of Kipf and Welling [2]. Mean and max graph pooling are concatenated with the mean failed-edge incidence rate. A two-layer head with sigmoid output predicts Equation (1). The model has three 48-dimensional hidden layers, dropout 0.15, and is trained using AdamW and mean squared error.

We evaluate three learned configurations. *Direct-full* uses all five features. *Direct-no-Fiedler* removes the third feature while preserving the architecture and split. The residual model predicts a bounded correction r_θ to the analytical estimate:

$$(4) \quad \hat{y}_{\text{res}}(G, S) = \text{clip}(\hat{y}_{\text{spec}}(G, S) + \tanh r_\theta(G, S), 0, 1).$$

This distinguishes a Fiedler coordinate used as a node feature from the analytical estimate used as an explicit inductive prior.

4. EXPERIMENTAL DESIGN

4.1. Synthetic spatial graphs. Random geometric graphs place nodes in the unit square and connect pairs within a radius. They are a basic spatial-network model rather than a faithful road generator. We adapt the radius until a connected graph with 35–65 nodes is obtained. Edge conductance is set to inverse Euclidean length. For each base graph we sample 20 scenarios, removing between one and eight edges.

The full benchmark contains five independent runs of 2,000 scenarios. Within each run, 70%, 15%, and 15% of the scenarios are assigned to training, validation, and test sets by *base graph identifier*. Consequently no damaged version of a training graph appears in the test set. Each run has 1,400 training, 300 validation, and 300 test scenarios, for a pooled total of 1,500 held-out predictions.

TABLE 1. Synthetic held-out performance, mean $\pm$ standard deviation over five seeds. Lower is better for MAE/RMSE; higher is better for R^2 and ρ .

Method	MAE	RMSE	R^2	Spearman ρ
Direct GCN	0.0656 ± 0.0132	0.1492 ± 0.0273	0.5464 ± 0.0952	0.6279 ± 0.0571
GCN without Fiedler	0.0694 ± 0.0133	0.1517 ± 0.0351	0.5372 ± 0.0910	0.6308 ± 0.0542
Residual spectral-GCN	0.0390 ± 0.0099	0.1083 ± 0.0320	0.7633 ± 0.0759	0.8764 ± 0.0375
Spectral first order	0.0652 ± 0.0243	0.2090 ± 0.0508	0.1331 ± 0.1222	0.9507 ± 0.0172

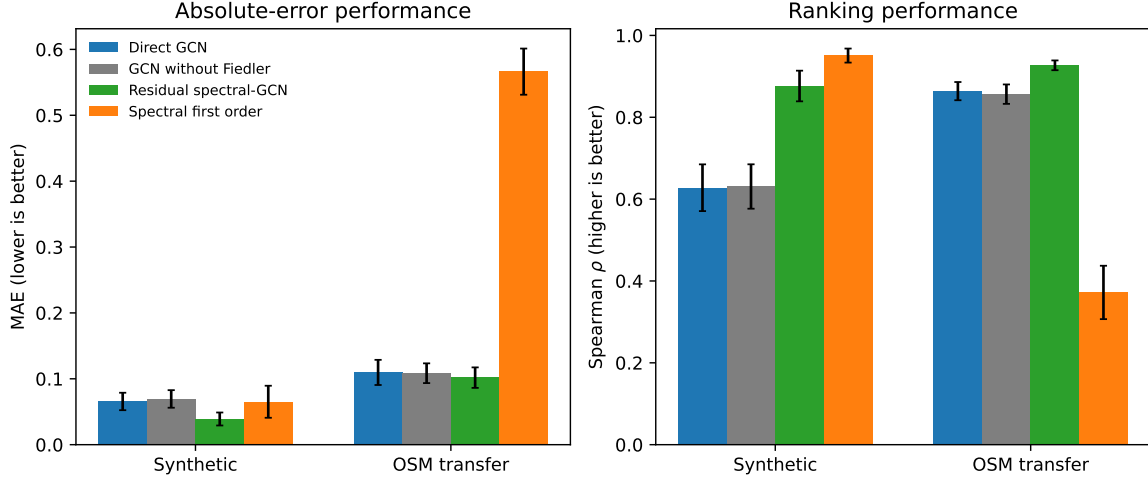


FIGURE 1. Synthetic and zero-shot OSM transfer. Error bars are standard deviations over five seeds.

4.2. Evaluation. We report mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2), and Spearman rank correlation ρ . The constant baseline predicts the training-target mean. Mean and sample standard deviation are computed over the five seeds (11, 22, 33, 44, 55), treating the seed rather than an individual correlated scenario as the independent replicate.

4.3. OpenStreetMap transfer. On September 1, 2026, OSMnx [3] was used to download the drivable network within 650 m of (10.7626, 106.6822), near the University of Science, VNU-HCM. The directed multigraph was simplified, converted to an undirected simple graph, and restricted to its largest connected component. The result has 163 nodes and 221 edges. Conductance is $100/\text{length}_m$ and coordinates are min-max normalized. For each seed, 300 independent random edge-disruption scenarios are generated. No OSM scenario is used for training, early stopping, or model selection: this is zero-shot transfer. Map data are credited to OpenStreetMap contributors under the Open Database License.

5. RESULTS

The ablation increases MAE by only 0.0038 on average and has essentially no effect on rank correlation. The node-level Fiedler coordinate is therefore not a dominant direct-GCN input. Residual learning reduces MAE by 40.6% and RMSE by 27.5% relative to direct GCN, while increasing ρ by 0.249. The standalone spectral approximation retains the best synthetic ranking but has much larger squared error.

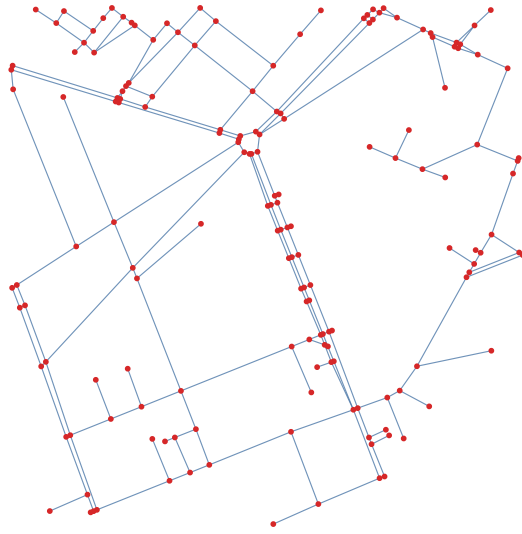


FIGURE 2. The 163-node, 221-edge OSM validation network near HCMUS after preprocessing. Map data copyright OpenStreetMap contributors.

On OSM transfer, direct GCN reaches 0.1097 ± 0.0191 MAE and 0.8639 ± 0.0223 rank correlation. Removing Fiedler changes MAE by only -0.0012 on average. Residual learning improves MAE to 0.1018 ± 0.0155 and rank correlation to 0.9269 ± 0.0119 . In contrast, the standalone first-order approximation deteriorates to 0.5662 ± 0.0350 MAE and negative R^2 . The sparse OSM topology lies outside the regime where a finite multi-edge deletion is well represented by a first-order sum; learning a correction recovers much of its ranking information.

6. DISCUSSION

The experiment answers a narrower question than real road resilience. It measures structural connectivity, not travel demand, capacity, delay, or recovery time. Moreover, random edge deletion is not a flood model. Flooded links are spatially correlated and depend on elevation, drainage, rainfall, and road geometry. Accordingly, the present results must be described as “disruption scenarios”, not flood predictions.

The evaluation exposes a useful methodological lesson. Randomly splitting all scenarios would allow damaged variants of one base graph in both training and test sets, producing optimistic estimates. Graph-level splitting should be mandatory in this setting. A direct GCN uses local message passing and struggles to rank a global eigenvalue response. Equation (4) combines the high-ranking spectral prior with a nonlinear correction, producing the most balanced method. Its OSM success is zero-shot, but it is based on one small geographic area and should not be interpreted as city-wide validation.

7. LIMITATIONS AND NEXT EXPERIMENTS

The following extensions remain necessary:

- (1) add spatially clustered failures and test out-of-range graph sizes;
- (2) measure inference time under a fixed batching and hardware protocol;
- (3) repeat OSM transfer over several areas and network densities;
- (4) only after obtaining a licensed hazard layer, study flood-conditioned failure probabilities.

No hypothesis test is emphasized with only five independent seeds. The current effect sizes and standard deviations are more informative than treating 1,500 dependent test scenarios as independent replicates.

8. CONCLUSION

Residual spectral-GCN substantially improves direct GCN on synthetic graphs and provides the best balance of error and ranking under zero-shot OSM transfer. The Fiedler-feature ablation shows that this gain comes from the explicit analytical prior rather than merely exposing a spectral node feature. The standalone first-order approximation is strong in-distribution but fails on the sparser OSM network, making the hybrid correction scientifically meaningful. Claims about urban flooding or policy must still wait for hazard data and multi-area validation.

REPRODUCIBILITY STATEMENT

The complete generators, spectral baseline, three learned configurations, graph-disjoint split, cached OSM GraphML, aggregation scripts, and generated tables are provided with the project. The runs used Python 3.13.11, CPU-only PyTorch with one compute thread, 2,000 synthetic and 300 OSM scenarios per seed, at most 100 epochs, and fixed seeds 11, 22, 33, 44, and 55.

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